



Republic of the Philippines
City of Malolos, Bulacan
BULACAN STATE UNIVERSITY
College of Information and Communications Technology



Management
System
ISO 9001:2015



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RODRIGUEZ, MILES J.

BSIT 2H-G1

Array Deletion Activities

Instructions:

1. Use a class name that matches the program and includes your full name.

□ Example: Prob01_CountTraversal_JuanDelaCruz
2. Type and run the program in your IDE or Notepad/CMD.
3. Paste both your code and the program output into this template.
4. Ensure that all programs are running successfully.

Problem 1: Delete Element at Specific Index

Problem Statement: Write a program that stores the numbers {10, 20, 30, 40, 50} in an array and Delete the element at index 2 from the array.

Program/Code:

```
public class Prob1_DeleteElementAtSpecificIndex_MilesRodriguez {  
    public static void main(String[] Koi) {  
        int[] numbers = {10, 20, 30, 40, 50};  
        int size = numbers.length;  
  
        System.out.println("Array Before Deletion: ");  
        for (int elements : numbers) {  
            System.out.print(elements + " ");  
        }  
  
        //Deleting Element on Index 2 (3rd element)  
        for (int i = 2; i < size - 1; i++) {  
            numbers[i] = numbers[i + 1];  
        }  
  
        --size;  
  
        System.out.println("\nArray After Deletion: ");  
        for (int i = 0; i < size; i++) {  
            System.out.print(numbers[i] + " ");  
        }  
    }  
}
```

Data Structures and Algorithm

Prepared by: Mr. E. B. Estrada



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Expected Output:

```
Array after deletion:  
10 20 40 50  
Get User Input: Break
```

Output:

```
Array Before Deletion:  
10 20 30 40 50  
Array After Deletion:  
10 20 40 50 Press any key to continue . . .
```



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Problem 2: Delete First Occurrence of a Value

Problem Statement: Create a program that stores {5, 15, 25, 15, 35} in an array. Delete the first occurrence of the value 15 from the array.

Program/Code:

```
public class Prob2_DeleteFirstOccurrenceOfAValue_MilesRodriguez {
    public static void main(String[] Koi) {
        int[] numbers = {5, 15, 25, 15, 35};
        int size = numbers.length;

        System.out.println("Array Before Deletion: ");
        for (int elements : numbers) {
            System.out.print(elements + " ");
        }

        //Finding the position of the first 15
        int posOfFirst15 = 0;
        for (int finder = 0; finder < size; finder++) {
            if (numbers[finder] == 15) {
                posOfFirst15 = finder;
                break;
            }
        }

        //Deleting the first 15 and adjusting the array
        for (int i = posOfFirst15; i < size - 1; i++) {
            numbers[i] = numbers[i + 1];
        }

        --size;

        System.out.println("\nArray After Deletion: ");
        for (int i = 0; i < size; i++) {
            System.out.print(numbers[i] + " ");
        }
    }
}
```

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Expected Output:

```
Array after deletion:  
5 25 15 35
```

Output:

```
Array Before Deletion:  
5 15 25 15 35  
Array After Deletion:  
5 25 15 35 Press any key to continue . . .
```



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Problem 3: Delete Last Element

Problem Statement: Create a program that stores {100, 200, 300, 400, 500} in an array. Delete the last element of the array.

Program/Code:

```
public class Prob3_DeleteLastElement_MilesRodriguez {  
    public static void main(String[] Koi) {  
        int[] numbers = {100, 200, 300, 400, 500};  
        int size = numbers.length;  
  
        System.out.println("Array Before Deletion: ");  
        for (int elements : numbers) {  
            System.out.print(elements + " ");  
        }  
  
        //Deleting the last element  
        --size;  
        for (int i = size; i < size; i++) {  
            numbers[i] = numbers[i + 1];  
        }  
  
        System.out.println("\nArray After Deletion: ");  
        for (int i = 0; i < size; i++) {  
            System.out.print(numbers[i] + " ");  
        }  
    }  
}
```



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Expected Output:

```
Array after deletion:  
100 200 300 400
```

Output:

```
Array Before Deletion:  
100 200 300 400 500  
Array After Deletion:  
100 200 300 400 Press any
```



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Reflection Questions:

1. How does the array change after deletion?

After deletion, the index to be deleted is just transferred to the end of the array and was just hidden or cannot be reached during printing. The array’s length cannot actually be shortened or decreased as arrays do have a fixed size.

2. Why do we shift elements when deleting by index or value?

We shift elements when deleting to let the element reach the end of the array and then technically delete it or hide it from being printed.

3. Why do we need to shift elements when deleting from the middle of the array, but not when deleting the last element?

We shift our elements to the left when a deletion was performed from the middle of the array to prevent an index from being empty, and shift the element to be deleted to the end of the array. On the other hand, there’s no need to shift elements when deleting the last element of the array, we can just simply hide the last element to prevent it from being printed.

Rubrics for Evaluation:

Criteria	Excellent (5)	Good (4)	Fair (3)	Needs Improvement (2)	Poor (1)
Correctness of Code	Runs without errors, output matches expected	Minor errors but mostly correct	Runs with errors but partially correct	Major errors, incorrect output	Code does not run
Understanding of Deletion	Clearly explains logic and effect	Explains with minor gaps	Basic explanation, limited detail	Minimal explanation	No explanation



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Output Verification	Output fully matches expected	Output matches with minor issues	Output partially matches	Output incorrect	No output shown
Reflection Answers	Complete, insightful, well-explained	Mostly complete, minor gaps	Basic answers, limited detail	Minimal answers	No answers
Submission (Code + Screenshot)	Code + output + screenshot submitted on time	Code + output submitted, screenshot missing	Code only submitted	Incomplete submission	No submission